$3x$ = Integer

$x = 1/3$

$3 \times \frac{1}{3}$ = Integer

So $x = 1/3$ is not an integer.

Hence, statement 1 and statement 2 both are insufficient.

20. The increase in the price of a certain item was 25%. Then the price was decreased by 20% and then again increased by 10%. What is the resultant increase in the price ?

(a) 5% (b) 10%

(c) 12.5% (d) 15%

Ans. (b) : According to the question, let the initial price of the item.

$$\text{Final price} = 100 \times \frac{125}{100} \times \frac{80}{100} \times \frac{110}{100}$$

$$= 110$$

Resultant increase % = 110 – 100 = 10%

Directions for the following 3 (three) items:

Read the following two passages and answer the items that follow the passages. Your answers tothese items should be based on the passages only.

Passage -1

In some places in the world, the productivity of staples such as rice and wheat has reached a plateau. Neither new strains nor fancy agrochemicals are raising the yields, Nor is there much unfarmed land left that is suitable to be brought under the plough. If global temperature continues to rise, some places will become unsuitable for farming. Application of technology can help overcame these problems. Agricultural technology is changing fast. Much of this change is brought about by affluent farmer in the West/Americans. Techniques developed in the West are being adapted in some place to make tropical crops more productive. Technology is of little use of it is not adapted. In the developing world, that applies as much to existing farming techniques as it does to the latest advances in genetic modification. Extending to the smallholders and subsistence farmers of Africa and Asia the best of today's agricultural practices, in such simple matters as how much fertilizers to apply and when, would lead to a greatly increased availability of food roads and storage facilities, at allow for the carriage of surpluses to markers and reduce wastage.

21. Based in the above passage, the following assumptions have been made:

1. Development of agricultural technology is confined to developed countries

2. Agricultural technology is not adapted in developing countries

Which of the above assumption is/are valid?

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

Ans. (d) : It is nowhere mentioned in the passage that agricultural technology development is limited to developed countries only. Nor is it said that agricultural technology has not been adopted in developing countries. Therefore, neither statement (1) nor statement (2) is valid.

22. Based on the above passage, the following assumptions have been made:

1. Poor countries need to bring about change in their existing farming techniques

2. Developed countries have better infrastructure and they waste less food

Which of the above assumption is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : Based on the above passage the following assumptions have been made–

– Poor countries need to bring about change in their existing farming techniques.

– It is not mentioned anywhere in the passage that developed countries have better infrastructure and they waste less food. Therefore, only assumption 1 is valid.

23. Based on the above passage, the following assumption have been made:

1. Growing enough food for future generations will be a challenge

2. Corporate farming is a viable option for food security in poor countries

Which of the above assumptions is/are valid

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

Ans. (d) : Based on the above passage, both of the assumptions i.e. neither 1 nor 2 is valid. Therefore, option (d) is the correct answer.

24. The letter A, B, C, D and E are arranged is such a way that there are exactly two letters between A and E. How many such arrangements are possible ?

(a) 12 (b) 18

(c) 24 (d) 36

Ans. (c) : According to the question,

A	—	—	E	— = 6
—	A	—	—	E = 6
E	—	—	A	— = 6
—	E	—	—	A = 6
				= 24

So total possible arrangements will be 24.

25. Consider the Question and two statements given below:

Question : Is Z brother of X ?

Statement-1 : X is a brother of Y and Y is brother of Z.

Statement-2 : X, Y and Z are siblings.

Which one of the following is correct in respect of the Question and the Statement

(a) Statement-1 alone is sufficient to answer the Question

(b) Statement-2 alone is sufficient of answer the Question

(c) Both Statement-1 and Statement-2 are sufficient to answer the Question

(d) Both Statement-1 and Statement-2 are not sufficient to answer the Question

Ans. (d) : From, statement 1 and 2

$$X \xleftarrow{\text{Brother}} Y \xleftarrow{\text{Brother}} Z$$

Here, it is not clear that Z is men or women.

So, both statement 1 and statement 2 are not sufficient to answer the question.

26. On one side of a 1.01 km long road, 101 plants are planted at equal distance from each other. What is the total distance between 5 consecutive plants ?

(a) 40 m (b) 40.4 m

(c) 50 m (d) 50.5 m

Ans. (b) : Distance between two plants

$$= \frac{1.01 \times 1000}{100} = 10.1 \text{ meter}$$

Distance between 5 consecutive plants

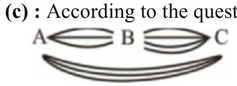
$= 10.1 \times 4 = 40.4$ meter.

27. A, B and C are three places such that there are three different roads from A to B, four different roads from B to C and three different roads from A to C. In how many different ways can one travel from A to C using these roads ?

(a) 10 (b) 13

(c) 15 (d) 36

Ans. (c) : According to the question,



So the total number of ways to go from A to C = (3 × 4) + 3 = 15

28. A has some coins. He gives half of the coins and 2 more a B. B gives half of the coins and 2 more to C. C gives half of the coins and 2 more to D. The number of coins d has now, is the smallest tow digit number. How many coins does A have is the beginning?

(a) 76 (b) 68

(c) 60 (d) 52

Ans. (d) : Let 'A' have number of coins = $8x$

B have number of coins = $4x + 2$

C have number of coins = $2x + 1 + 2$

$$\text{D have number of coins} = x + \frac{3}{2} + 2$$

$$= x + \frac{7}{2}$$

$$\because x + \frac{7}{2} = 10$$

$$\because \text{ A have number of coins} = 8 \times \frac{13}{2} = 52$$

29. In the series AABABCABCDABCDABCDE...., which letter appears at the 100th place?

(a) G (b) H

(c) I (d) J

Ans. (c) : A AB ABC ABCD ABCDE…….

1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 11+12 +13 = 91

A B C D E F G H I J K L M N

92 93 94 95 96 97 98 99 100

So, 'I' will come at 100th place.

30. Three persons A, B and C are standing in a queue not necessarily in the same order. There are 4 persons between A and B and 7 persons between B and C. If there are 11 persons ahead of C and 13 behind A, what could be the minimum number of persons in the queue?

(a) 22 (b) 28

(c) 32 (d) 38

Ans. (a) :



Total minimum number of person = 10 + 12 = 22

Directions for the following 3 (three) items:

Read the following two passages and answer the items that follow the passages. Your answers tothese items should be based on the passages only.

Passage–1

Natural selection cannot anticipate future environments on the earth. Therefore, the set of existing organisms can never be fully prepared for environmental catastrophes that await life. An outcome of this is the extinction of those species which cannot overcome